

PARTH DHARMENDRA PRAJAPATI

parthdpraja@gmail.com | +91 9152015908 | [LinkedIn](#) | [Portfolio](#) | OCI Certified Generative AI Professional | AWS Certified AI Practitioner

EDUCATION

- University of Southern California** Los Angeles, California
Master of Science in Computer Science (CGPA: 3.52 / 4.0) August 2023 - May 2025
- Relevant Courses: AI, Machine Learning for Data Science, Analysis of Algorithms, Database Systems, Web Technologies
- University of Mumbai** Mumbai, India
Bachelor of Engineering in Information Technology (CGPA: 9.19 / 10) August 2019 - May 2023
- Relevant Courses: Data Structures and Algorithms, Web Development, Operating Systems, Big Data Analytics

SKILLS

- Programming languages:** Python, SQL, JavaScript, Java, C
- AI/ML Libraries:** PyTorch, TensorFlow, Scikit-learn, Keras, NumPy, Matplotlib, Pandas, Hugging Face
- GenAI & LLM:** Transformers, LangChain, TRL (Transformer Reinforcement Learning), RAG, OCI GenAI, OpenAI API
- Cloud & Web:** AWS, Oracle Cloud Infrastructure (OCI), React, Node.js, MongoDB, PostgreSQL, Flask, Render, Docker

PROFESSIONAL EXPERIENCE

- University of Southern California** Los Angeles, California
Research Assistant October 2025 – Present
- Conducting research on **machine learning based pattern recognition** of Indus Valley scripts from historical symbols
 - Implementing a language translation model using **Transformer architecture** to create technical demonstrations for the AI curriculum
 - Designing **path optimization algorithms** for Geographical Information Systems to visualize and improve emergency response routes
 - Documenting technical specifications and architectural decisions to ensure reproducibility and support the iterative development of the model frameworks
- Vayu Technology Corp.** Los Angeles, California
Data Science Intern August 2024 – April 2025
- Designed and implemented **machine learning algorithms** for large scale **time-series sensor data**
 - Developed **production ready Python modules** for multiple activity features using **time-series signal processing**, successfully integrated into the company's core application
 - Built a robust **threshold-based model for calibration checks** across multiple distributed body segment sensors, **reducing error rates to less than 10%**
 - Created and refined prompts for an **LLM based applications**, improving **response consistency and evaluation quality**
 - Collaborated in a **cross-functional team, documenting designs** and delivering maintainable Python code
- University of Southern California** Los Angeles, California
Lead Course Producer (ISE 583: Enterprise-Wide Information Systems) January 2025 – May 2025
- Led course operations for an **enterprise systems program, coordinating cross-functional teams** and managing 59 students
 - Evaluated assignments, delivering timely, feedback driven assessments to enhance learning outcomes and **ensure quality standards**
 - Conducted ERP simulations, **performing testing and validation** of assignments while reinforcing enterprise architecture concepts
- University of Southern California** Los Angeles, California
Teaching Assistant (ITP 259: Basics of Artificial Intelligence) August 2024 – December 2024
- Assisted in teaching core **Artificial Intelligence concepts**, graded assignments and mentored to 30 students
 - Supported students with course material clarification and conducted open lab sessions to facilitate **hands-on learning**
 - Performed **testing and debugging** of assignments, resolving student queries to ensure conceptual clarity

ACADEMIC PROJECTS

- SmartMedAI: Fine-tuning MedicalQA-based LLM using Reinforcement Learning from AI Feedback (RLAIF)**
- Fine-tuned a MedicalQA-based LLM** using LoRA, which optimized training efficiency and enabled domain-specific adaptation
 - Built a preference dataset using **RLAIF** with a secondary LLM evaluating open-ended medical responses for DPO
 - Developed a **Direct Preference Optimization (DPO) pipeline** to align the model for clinically accurate and relevant responses, achieving 80.4% accuracy and 0.568 BLEURT Score
- AI News Aggregator**
- Automated daily **AI news digests** by scraping YouTube & RSS feeds with **LLM-powered summarization** delivered to users' inboxes
 - Built an **OpenAI-driven relevance-ranking pipeline using PostgreSQL** to surface personalized AI industry updates with less noise
 - Deployed a **containerized end-to-end GenAI app** to production using **Docker** and **Render**
- Stock Analytics Web Application**
- Developed a **full-stack web application** using **React, Node.js, and Express.js**, integrating real time stock data via **Finnhub API**.
 - Implemented features dynamic Highcharts and **social media sharing**, enhancing user interaction and **data visualization**
 - Implemented **MongoDB for data management** and deployed the application on **AWS**, specifically using **Amazon EC2 instances**
- Image Classification using Transfer Learning**
- Developed a **multi-class image classifier** using transfer learning with **ResNet50, ResNet101, EfficientNetB0, and VGG16 models**
 - Applied frozen feature extraction and **advanced data augmentation** (cropping, rotation, flipping) to **address limited data**
 - Achieved 91.8% accuracy with EfficientNetB0, outperforming deeper models using TensorFlow and Keras frameworks